

BEDFORD, United KingdomWMO: **035600**Lat: **52.2269N**Lon: **0.4653W**Elev: **276**StdP: **14.55**Time Zone: **0.00 (EUW)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.3	27.8	22.0	16.7	29.8	24.4	18.7	31.3	31.0	48.1	27.9	47.9	7.4	60	0.623

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	80.9	65.3	76.7	63.8	73.3	62.4	67.2	76.8	65.3	73.7	63.8	71.3	9.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.9	89.8	70.2	62.2	84.6	68.3	60.7	80.1	66.7	31.8	76.9	30.3	74.1	29.1	71.4	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.5	22.6	20.3	DB	20.6	86.6	4.5	3.5	17.4	89.1	14.8	91.1	12.3	93.1	9.1	95.6	(4)
(5)			WB	20.2	71.0	4.3	3.4	17.2	73.5	14.7	75.5	12.3	77.4	9.2	79.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.5	40.1	40.7	43.6	47.9	53.1	58.6	62.7	62.6	58.1	51.9	45.0	40.7	(6)	
	DBStd	9.82	5.99	5.97	5.46	5.36	5.26	4.72	4.56	4.56	4.44	5.15	5.61	6.65	(7)	
	HDD50	1405	310	263	206	100	27	1	0	0	1	38	165	294	(8)	
	HDD65	5373	773	680	663	512	368	199	102	104	213	407	599	753	(9)	
	CDD50	1581	2	3	8	38	124	260	394	391	243	96	16	6	(10)	
	CDD65	75	0	0	0	0	0	8	31	30	5	1	0	0	(11)	
	CDH74	672	0	0	0	3	9	93	263	258	42	4	0	0	(12)	
	CDH80	140	0	0	0	0	0	15	55	65	5	0	0	0	(13)	
(14) Wind	WSAvg	10.2	11.5	11.7	11.3	10.2	9.9	8.9	8.8	9.0	9.4	10.3	10.3	10.9	(14)	
(15) Precipitation	PrecAvg	23.6	2.0	1.5	1.3	1.8	1.8	1.9	2.0	2.2	2.1	2.4	2.4	2.0	(15)	
	PrecMax	36.9	4.3	3.1	3.3	5.2	4.7	4.4	4.2	5.2	5.9	5.2	5.2	3.7	(16)	
	PrecMin	15.4	0.4	0.1	0.1	0.1	0.2	0.3	0.6	0.1	0.4	0.5	0.9	0.6	(17)	
	PrecStd	4.9	0.9	0.8	0.8	1.3	1.1	1.3	1.0	1.2	1.4	1.1	1.0	0.9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	57.7	61.3	71.0	75.1	82.2	85.9	86.3	79.3	69.5	60.1	56.9	(19)	
		MCWB	52.4	52.6	52.3	58.1	62.1	67.1	67.3	67.1	63.1	61.4	57.2	53.6	(20)	
	2%	DB	53.3	53.8	57.6	64.6	71.2	76.1	80.4	81.0	73.2	64.4	57.8	54.9	(21)	
		MCWB	50.7	49.9	50.9	55.3	61.1	64.0	65.3	65.4	62.3	58.6	54.9	52.4	(22)	
	5%	DB	51.2	51.5	54.8	60.8	67.2	72.1	76.3	75.7	69.4	61.7	55.6	52.9	(23)	
		MCWB	48.6	48.1	49.6	52.8	58.5	61.8	63.9	63.5	60.2	57.3	52.8	50.5	(24)	
	10%	DB	49.1	49.6	52.6	57.5	63.1	68.6	72.8	72.1	66.5	59.8	53.5	50.7	(25)	
		MCWB	46.6	46.6	48.0	50.9	56.0	59.8	62.1	61.7	58.7	56.0	51.0	48.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.0	53.3	54.0	59.8	65.7	69.3	69.3	70.2	66.9	62.4	57.7	54.4	(27)	
		MCDB	54.7	56.5	58.5	67.9	71.2	79.9	80.5	80.2	74.6	67.3	59.2	56.1	(28)	
	2%	WB	51.1	50.5	52.1	56.3	62.0	65.3	67.0	66.5	64.0	60.0	55.6	52.8	(29)	
		MCDB	53.1	53.0	56.1	63.2	69.0	73.2	76.6	76.4	70.4	63.2	57.5	54.7	(30)	
	5%	WB	48.9	48.8	50.5	54.0	59.6	63.0	65.2	64.8	62.1	58.3	53.4	50.9	(31)	
		MCDB	50.9	51.1	54.1	59.2	65.5	70.1	73.9	73.3	67.4	61.0	55.2	52.7	(32)	
	10%	WB	46.9	47.1	48.7	51.7	57.2	61.0	63.4	63.1	60.3	56.6	51.4	48.7	(33)	
		MCDB	48.9	49.4	52.2	56.9	62.2	67.4	71.2	70.2	64.8	59.4	53.2	50.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	10.7	13.2	15.8	16.1	16.4	17.0	16.8	15.3	12.6	10.2	8.9	(35)	
		MCDBR	11.1	12.0	16.3	20.7	22.4	22.5	23.4	22.8	20.0	13.8	10.8	10.0	(36)	
	5% WB	MCWBR	10.4	9.5	11.0	12.5	13.5	12.2	10.4	10.1	10.3	9.3	9.4	9.4	(37)	
		MCWBR	10.9	11.7	15.0	18.9	19.6	20.0	21.2	19.8	17.0	12.7	10.8	10.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.322	0.339	0.372	0.396	0.401	0.402	0.400	0.389	0.370	0.352	0.326	0.312	(40)	
		taud	2.371	2.388	2.287	2.231	2.255	2.274	2.289	2.342	2.401	2.436	2.418	2.352	(41)	
	Ebn at Noon		213	246	258	265	270	270	268	265	257	237	211	194	(42)	
		Edh at Noon	19	25	33	40	41	40	39	35	30	23	18	16	(43)	
(44) All-Sky Solar Radiation	RadAvg		245	462	802	1232	1513	1633	1564	1294	971	565	299	198	(44)	
	RadStd		21	57	100	136	97	161	166	95	86	49	33	21	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (91 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page